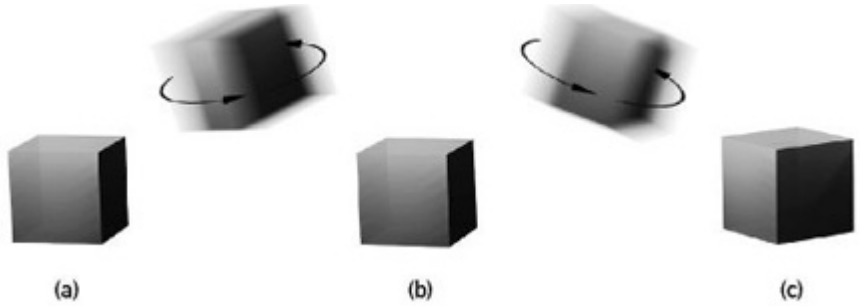


Figure 8.1 If a cube, as in **(**a**),** is rotated by 90 degrees, or multiples thereof, around axes passing through any of its faces, it looks unchanged, as in **(**b**).** But any other rotations can be detected, as in (c).

All these are examples of symmetries of objects *in* space. The symmetries underlying the known laws of physics are closely related to these, but zero in on a more abstract question: what manipulations—once again, real or imagined—can be performed on you or on the environment that will have absolutely no effect on the *laws* that explain the physical phenomena you observe? Notice that to be a symmetry, manipulations of this sort are not required to leave your observations unchanged. Instead, we are concerned with whether the laws governing those observations—the laws that explain what you see before, and then what you see after, some manipulation—are unchanged. As this is a central idea, let's see it at work in some examples.

Imagine that you're an Olympic gymnast and for the last four years you've been training diligently in your Connecticut gymnastics center. Through seemingly endless repetition, you've got every move in your various routines down perfectly—you know just how hard to push off the balance beam to execute an aerial walkover, how high to jump in the floor exercise for a double-twisting layout, how fast to swing on the parallel bars to launch your body on a perfect double-somersault dismount. In effect, your body has taken on an innate sense of Newton's laws, since it is these very laws that govern your body's motion. Now, when you finally do your routines in front of a packed audience in New York City, the site of the Olympic competition itself, you're banking on the same laws holding, since you intend to perform your routines exactly as you have in practice. Everything we know about Newton's laws lends credence to your strategy. Newton's laws are not specific to one location or another. They don't work one way in Connecticut and another way in